Modeling Anthropogenic CO2 Accumulation: A Bayesian Comparison of Linear and Accelerating Trends

Álvaro Méndez Rodríguez de Tembleque[a]

[a] *Universita degli Studi di Padova, Department of physics and astronomy "Galileo Galilei", City, 12345*

Abstract

The Mauna Loa Observatory is a premier baseline site for monitoring historical and current atmospheric carbon dioxide (CO2) concentrations. This report investigates whether the long-term increase in atmospheric CO2 is best described by a constant linear increase or an accelerating quadratic trend. Using monthly average CO2 data from the NOAA Global Monitoring Laboratory, we address missing values and compute annual averages to isolate the long-term trend from seasonal variations. We perform a Bayesian analysis to estimate the parameters of both a linear model and a quadratic model, with a specific focus on the acceleration parameter. The two models are then evaluated and compared using an approximate Bayes factor to determine which model better captures the underlying trend dynamics. Finally, utilizing the best-fitting model, we project the annual atmospheric CO2 concentrations for the next decade, providing predictive estimates complete with Bayesian credible bands to quantify uncertainty.

---

1. Introduction

See code 1 in Appendix for the implementation.

*Corresponding author

Email address: `alvaro.mendezrodriguezdetembleque@studenti.unipd.it` (Álvaro Méndez Rodríguez de Tembleque)

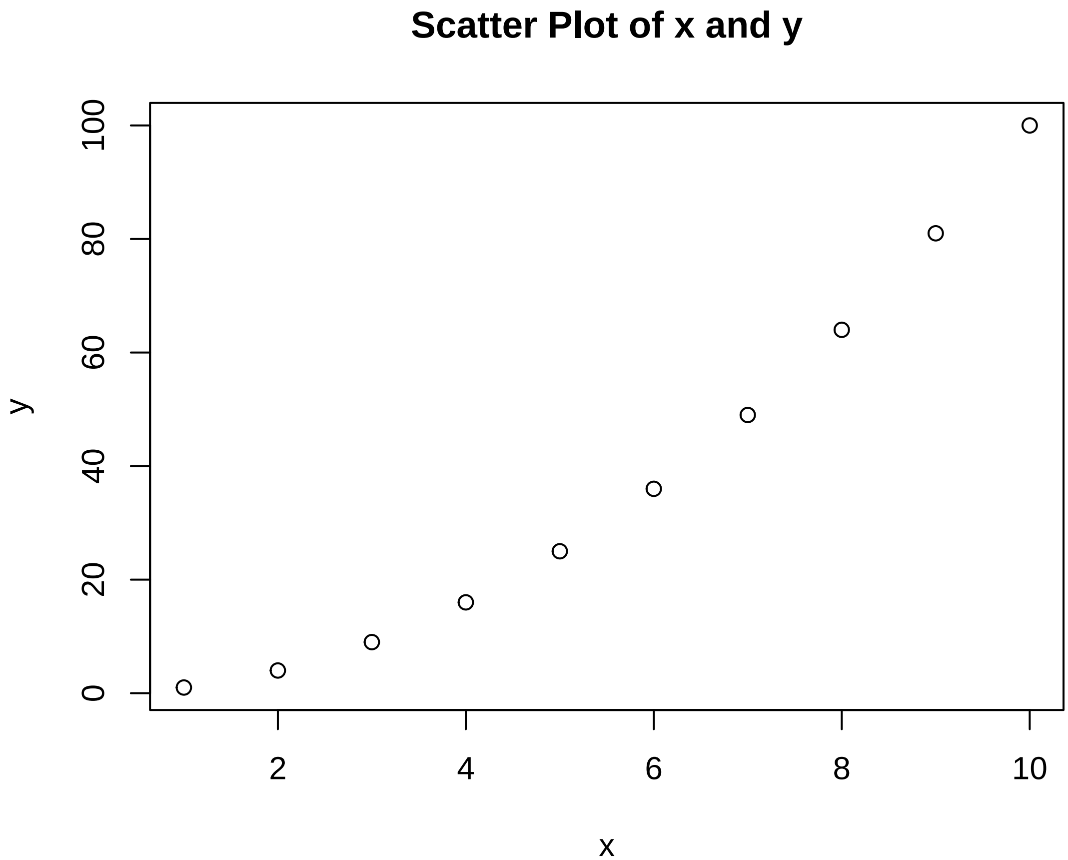

Figure 1: Scatter plot of the binary outcomes and the logistic regression curve

1.1. Scientific Context

1.2. Objectives

2. Data Description and Preprocessing

2.1. Dataset Overview

2.2. Data Cleaning

2.3. Aggregation

2.4. Exploratory Visualization

3. Bayesian Methodology and Models

3.1. Model 1: Linear Trend

3.2. Model 2: Quadratic Trend

3.3. Inference Strategy

4. Results and Inference

4.1. Parameter Estimation

4.2. Model Comparison

5. Predictive Forecasting

5.1. Ten-Year Projections

5.2. Uncertainty Quantification

6. Conclusion

6.1. Summary of Findings

6.2. Limitations and Future Work

7. References

7.1. Appendix: Source code

```

# This is an example of a test file for EDA (Exploratory Data Analysis)

# Load necessary libraries
x <- 1:10
y <- x^2
# Create a scatter plot
plot(x, y, main = "Scatter␣Plot␣of␣x␣and␣y", xlab = "x", ylab = "y")
```

Listing 1: Plotting the data